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Next-Generation Computer Vision in Veterinary Medicine: A Study on Canine Ophthalmology

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Keywords

ai in veterinary medicine, canine ophthalmology, computer vision, computing and processing, gpt-4o, grounding sam, image segmentation, machine learning, medical image analysis, ocular imaging, signal processing and analysis, u-net, veterinary diagnostics

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Abstract—This study aims to evaluate the application of three computer vision methodologies for canine ophthalmology: U-Net and an enhanced U-Net model, GPT-4 and its latest iteration GPT-4o, and Grounding Dino in combination with Segment Anything (SAM). The purpose is to address the limitations of existing methods and enhance diagnostic accuracy and efficiency in veterinary ophthalmology. The research employs an empirical comparison of the three methodologies, focusing on their segmentation effectiveness and interpretative accuracy. The U-Net model is assessed for its precision in medical image segmentation, GPT-4 and GPT-4o are focused on their interpretative analysis, and Grounding SAM for its innovative combination of image recognition and segmentation. The enhanced U-Net model demonstrated superior segmentation performance, accurately delineating ocular structures in canines. GPT-4o excelled in providing detailed interpretative textual descriptions akin to a specialist's assessment, though it fell short in segmentation accuracy compared to U-Net. Grounding SAM, despite being less mature in development, showed promise in offering comprehensive analysis through its combined segmentation and contextual insights. Integrating these advanced computer vision techniques can significantly enhance diagnostic accuracy and efficiency in veterinary ophthalmology, particularly for diagnosing and monitoring canine ophthalmic conditions. This research underscores the transformative potential of these methodologies in veterinary practices, highlighting both their benefits and limitations. The findings suggest that a synergistic approach leveraging the strengths of each method could pave the way for significant innovations in veterinary and medical imaging.

Keywords— Canine ophthalmology, computer vision, U-Net, GPT-4o, Grounding SAM, image segmentation, veterinary diagnostics, AI in veterinary medicine, ocular imaging, machine learning, medical image analysis.

I. INTRODUCTION

Computer vision has significantly advanced the field of medicine, providing new methods for diagnosing and monitoring various medical conditions. The methods used in recent studies are driven by deep learning techniques using neural networks and transformers which supply satisfactory results in obtaining information about classification and certain object localization.

The problem of automatic diagnosis of eye diseases in dogs is challenging. In addition to the classic reasons that increase the complexity of training a machine learning model that refer to a small area of interest in an image that sometimes occupies only a few pixels, different camera positions and distance from the object, motion blur, changes in lighting, crowded background, occlusions and the like, additionally are

challenges related to significant differences in dog breeds. No less important is the fact that collecting images is not easy, because it is not easy sometimes even for the owner to calm down the sick dog and position him so that his sick eye can be recorded in such a way that all the symptoms can be seen. Illustrative examples Fig. 1 from the study [1], [2] include images of an American bulldog and a crossbred dog, which show the same diagnosis of eye disease with a pink-colored ocular protrusion from the medial canthus and symptoms of excessive tearing indicative of Cherry Eye in different dogs and illustrates some of the challenges that models for automatic segmentation and interpretation of eye disease must deal with.

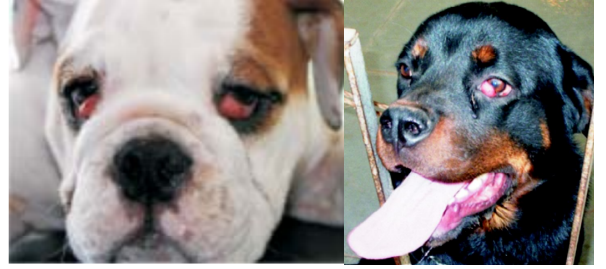


Fig. 1. Images of an American bulldog and crossbred dog with protrusion of pink colored mass from medial canthus of both eyes with inflammatory signs involving pain and epiphora suffering from Cherry Eye [39, 40].

The application of computer vision technologies has the potential to enhance the accuracy and efficiency of ophthalmological assessments. We assume that the integration of advanced computer vision techniques such as encoder-decoder and convolutional deep neural networks with the power of transformers can significantly improve diagnostic accuracy and efficiency in veterinary ophthalmology compared to using these methodologies independently.

This article focuses on semantical segmentation of medical symptoms and disease diagnosis in canine ophthalmology as shown in Fig. 2. The input consists of a dog's eye image, which undergoes segmentation and interpretation processes using: a commonly used deep neural network U-Net [3] and an enhanced U-Net model [4], multimodal large language models (LLM) GPT-4 [5], GPT-4o [6], and a Transformer-based model Grounding Dino [7] with Segment Anything (SAM) [8]. As an output, segmented symptoms along with the disease diagnosis are achieved.

This study aims to provide an empirical comparison of U-Net, Grounding SAM models, and GPT-4o methodologies individually and when used incrementally in a pipeline, considering their segmentation effectiveness and

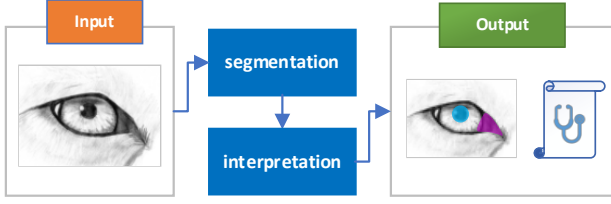


Fig. 2. Workflow diagram of canine ophthalmic condition analysis illustrates the process of analyzing a dog's eye image, where segmentation is performed using U-Net, Grounding SAM models and GPT-4o, followed by interpretation of the medical symptoms by GPT-4 and GPT-4o, resulting in a segmented image and a complete diagnostic report.

interpretative outputs. The results of the interpretation of the diagnosis are in the form of a text, so for their evaluation, to be as objective as possible, we used the metrics that are commonly used in the evaluation of the text, such as BERTScore [9], CLIPScore [10], BLEU [11], METEOR [12], ROUGE [13], and SPICE [14]. Also, the research considers the current problem of applying deep learning in the field of medicine, which mostly has a relatively small number of medical images with a diagnosis that is processed and annotated and ready for training models, which, on the other hand, need as much diverse data as possible to be able to achieve satisfactory performance.

The goals of the research are to contribute to the integration of computer vision in veterinary practices, especially in the diagnostic processes of ophthalmology, and the main contributions can be summarized as:

- Usage of Stable Diffusion models for the generation of training dataset images with ophthalmology symptoms
- Proposed enhanced U-Net model for segmentation of medical symptoms in veterinary ophthalmology medicine
- Pipeline for segmentation and interpretation of canine eye disease images
- Segmentation results comparison provided by different variations of U-Net, Grounding SAM, and LLMs
- Evaluation of diagnosis provided by GPT-4 and GPT-4o using selected text similarity metrics

The paper's structure includes a related work where recent studies on the topic of medical image segmentation and interpretation are described. This is followed by methodology which defines the dataset preparation, image generation, and methods picked for segmentation and diagnosis prediction. Next is the detailed description of the experiment divided into segmentation and interpretation with the results. Finally, a discussion and conclusion showing strengths and weaknesses are provided with suggestions for further research.

II. RELATED WORK

The application of computer vision in veterinary medicine, specifically in canine ophthalmology, has been gradually researched in recent years [16], [17], [18].

Recent advancements in image segmentation have led to the development of various techniques, enhancing precision and efficiency. Notable methods include encoder-decoder networks like SegNet [19] and DeepLabv3+ [20], which utilize advanced architectures for better localization and boundary segmentation by incrementally recovering spatial

information. High-resolution networks such as HRNet [21] maintain detailed representations throughout the process, ensuring high spatial accuracy. Transformer-based models like Swin Transformer [22] and SegFormer [23] leverage self-attention mechanisms to capture both fine-grained details and global context. Additionally, the Dense Prediction Transformer (DPT) [24] employs dense feature extraction with vision transformers to improve segmentation quality by aggregating long-range dependencies. The Pyramid Scene Parsing Network (PSPNet) [25] is another notable example, employing a pyramid pooling module to capture multi-scale context, leading to improved segmentation performance across different scales. These methods represent the state-of-the-art in segmentation, each contributing unique strengths to the field. Despite these advancements, we decided to use The U-Net model which has been widely recognized for its effectiveness in medical image segmentation from the first time it was introduced till the recent studies where it still shows competitive results [26], [27]. Its architecture, designed with medical image analysis in mind, has proven to be highly precise in image segmentation tasks, especially in ophthalmology [28], [29], [30]. A study comparing U-Net with transformer-based approaches in medical image registration found that with only modest modifications, U-Net can outperform transformer-based architectures [31]. Another research introduced the U-Transformer network, which combines a U-shaped architecture for image segmentation with self/cross-attention from Transformers, overcoming the inability of U-Nets to model long-range contextual interactions and spatial dependencies [27]. Furthermore, U-Net variants such as the [29] have demonstrated enhanced performance in specific segmentation tasks by incorporating attention mechanisms.

Generative Pre-trained Transformer (GPT) models, such as GPT-4, compared to U-Net, offers a different approach. They utilize a decoder structure based on the Transformer architecture which utilizes self-attention mechanisms to weigh the importance of different words in a sentence. At the input to the network is typically text sequence that is tokenized and processed through network layers, where self-attention computes weighted sums of token embeddings. That hierarchical architecture of the network allows coherent text to be generated based on the context provided by the input data making it highly effective for various natural language processing tasks [32]. Although, the GPT-4 was designed for language generation tasks already has shown considerable potential in various domains, including veterinary medicine [33]. GPT-4 can extract patient data, generate progress notes, and potentially assist in diagnosing complex cases [34]. It is important to note that while GPT-4 can generate highly believable and coherent text in a wide range of contexts, it relies solely on statistical representations of language and lacks any understanding of the meanings behind its generated output [33]. Studies have explored the integration of language models like GPT-4 with medical data to provide interpretative analysis, showing promise in improving diagnostic workflows [35].

Grounding SAM (Segment Anything Model) is the youngest methodology that combines image recognition with zero-shot learning to deliver segmentation alongside interpretive insights. Grounding DINO is capable of zero-

shot detection of an object in the image, while SAM can convert bounding boxes of an object into instance segmentation masks. Research on zero-shot learning for medical image analysis indicates that such models can generalize well to new tasks without additional training [36]. Since the release of Grounding SAM, there have been several applied research on various medical images [37] particularly on liver tumor [38] and polyps [39] image segmentation.

In the realm of artificial intelligence and computer vision, generating synthetic data has become a powerful assistant to augment training datasets. There are two primary approaches to synthetic data generation: Rules-based and Generative AI methods. The rules-based method involves creating synthetic data based on predefined rules and algorithms. It relies on domain knowledge to simulate realistic data scenarios. While this approach can be effective for generating structured and controlled data [40], it may lack the variability and realism that generative models can provide. Rules-based methods are often used in environments where specific conditions and constraints need to be met [41], but they might not capture the full complexity of real-world data as effectively as generative AI techniques. Generative AI utilizes advanced machine learning models to create realistic synthetic data. Techniques such as Generative Adversarial Networks (GANs) [42] and diffusion models [43] fall under this category. These models are trained on real data to learn its distribution and generate new, realistic samples. Diffusion models, for instance, iteratively add and remove noise to create high-quality images from random noise [44]. One example is a Stable Diffusion model that iteratively denoises a latent variable to generate high-quality images. The model operates by learning a series of reverse steps, starting from pure noise and progressively refining the image through the application of learned noise patterns. The approach leverages the U-Net architecture for image synthesis and employs a variational autoencoder (VAE) to handle the latent representations, making it particularly suited for high-resolution image generation. This approach has proven effective in creating highly realistic and diverse images using conditional control [45], that can enhance the robustness and performance of AI models [46] in various applications, including training datasets for segmentation models [47].

In the following, we will experimentally examine the adaptability of the mentioned methodologies and the possibility of application in the field of canine ophthalmology to gain an insight into their advantages and limitations when applied in veterinary practice.

III. METHODOLOGY

A. Training dataset description and test dataset preparation

By design, to produce an effective U-Net model capable of segmenting desired objects it is necessary to train the model using a dataset. Datasets can be obtained from an available repository or custom datasets can be generated. Due to the specificity of the research custom dataset for training was applied which was introduced in [48].

The dataset includes a collection of 200 canine ocular images, of diverse dog breeds, each resized to 320x320 pixels and gathered from a veterinary medical encyclopedia,

medical articles with documented diagnoses and treatments, and veterinary clinics with appropriate permissions [49]. The images in the dataset were annotated according to four common medical symptoms (see Fig. 3 [48]):

- S1 - cloudiness or haziness of the cornea,
- S2 - sclera redness,
- S3 - excessive tears, and a
- S4 - colored mass protrusion in the corner of the eye.

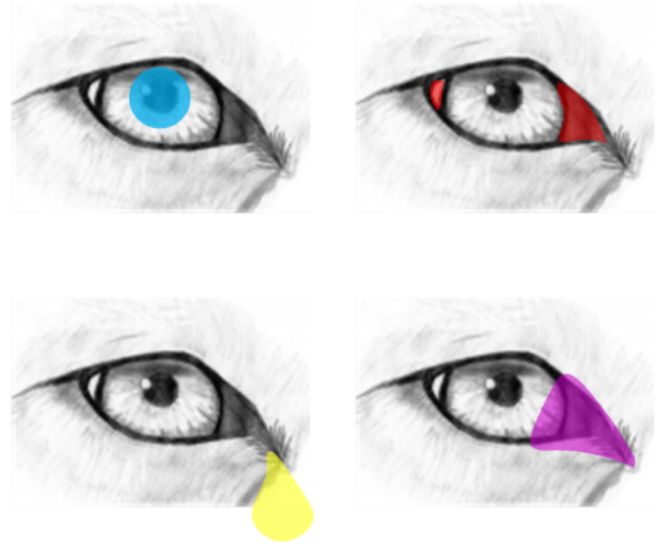


Fig. 3. In the top-left to bottom-right order: S1 - cloudiness or haziness of the cornea; S2 - sclera redness; S3 - excessive tears; and a S4 - colored mass protrusion in the corner of the eye.

Each image was associated with a one-channel mask stored in PNG format, annotating four distinct classes represented by varying pixel intensities. To ensure balanced representation across classes, images were carefully selected to address any overrepresentation of one class compared to others. Additionally, augmentation techniques such as image flipping, shifting, and rotation were applied to enhance the robustness of the dataset. In veterinary ophthalmology, the following diseases are commonly diagnosed using the mentioned symptoms: Cherry Eye, Glaucoma, Uveitis, Corneal Ulceration, and Bacterial Keratitis. Cherry Eye is evident when the gland of the third eyelid prolapses, resulting in a colored mass protrusion in the corner of the eye, potentially leading to dryness and irritation if untreated. Glaucoma manifests with increased intraocular pressure causing pain, sclera redness, haziness of the eye, and potentially vision loss if not promptly managed. Uveitis involves inflammation of the uveal tract, presenting symptoms such as sclera redness, pain, light sensitivity, and corneal cloudiness, often necessitating immediate veterinary care. Corneal Ulceration involves scratches or erosions on the corneal surface, indicated by excessive tears, sclera redness, and sometimes corneal cloudiness, requiring medical intervention to prevent infection and preserve vision. Bacterial Keratitis, an infection of the cornea, typically marked by corneal cloudiness, excessive tears, and ocular discomfort, demands antibiotic treatment to clear the infection and avert further damage to the eye. Based on the described diseases the test dataset was generated where diseases were grouped into five groups with five cases each resulting in 25 images. Before input into the respective models, images underwent preprocessing, which included

resizing to 320x320 pixels and normalization to enhance performance consistency. The final test dataset consists of close-up images of eye illness with the ground truth consisting of corresponding masks in PNG format where each class has a corresponding pixel value, along with the name of the diagnosed disease and symptoms. An example of the test dataset is shown in Fig. 5.



Fig. 4. Example images generated using Stable Diffusion with Cherry Eye disease in the lower right corner of a closer eye in various dog breeds.

B. Synthesized images dataset generation

In order to increase the amount of relevant medical data needed for proper model training, Stable Diffusion, a powerful generative model has been leveraged to create synthetic eye images of dogs with specific and various ophthalmic conditions exhibited by canines. Publicly available models generally provide satisfactory results, but they often lack the basic knowledge required for specific medical conditions. Additionally, public models tend to perform well with portraits of animals Fig. 4, but when close-ups of an eye are desired, the output is often related to human subjects. To ensure the synthetic images accurately depict the desired ophthalmic conditions, a custom LoRA (Low-Rank

Adaptation) [50], [51], a parameter-efficient fine-tuning method was trained using a subset of real images [48] of canine eye diseases. The LoRA model adapts the weights of the Stable Diffusion model by learning low-rank updates, which significantly reduces the computational cost and memory requirements. This adaptation allows the model to specialize in generating images with specific pathologies, such as corneal cloudiness, sclera redness, excessive tearing, and the presence of colored masses indicative of conditions like Cherry Eye and Glaucoma.

The synthesized dataset comprises images representing various stages and severities of ophthalmic conditions. Key steps in the dataset generation process include:

- Step 1: Definition of the specific ophthalmic conditions and symptoms to be generated: Cherry Eye, Glaucoma, Uveitis, Corneal Ulceration, and Bacterial Keratitis (these symptoms are described in Fig. 3).
- Step 2: Guiding the trained Stable Diffusion model in generating realistic images of diseased eyes with textual descriptions of the desired symptoms written in prompts like "eye with prolapsed eyelid gland" or "eye with red sclera" (see Fig. 6).
- Step 3: Evaluation of generated images to ensure clinical relevance and accuracy.
- Step 3.1: If the generated image did not resemble real eye diagnoses or had some kind of deficiency (unrealistic animal fur, abnormalities such as surplus ears; see Fig. 7 and Fig. 4), it was returned to Step 2. using additional conditions to refine the model results. Iterative feedback loops were employed until the model output was refined enough to faithfully simulate images from real-world conditions.
- Step 3.2: If the generated image is of satisfactory quality and matches real-world eye diagnosis images, place the image in the model's training set.

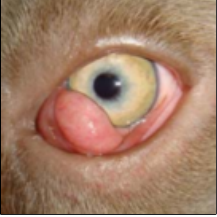
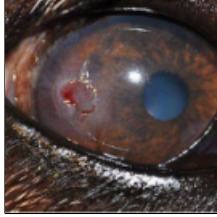
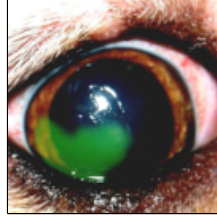
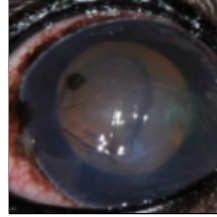
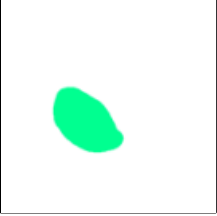
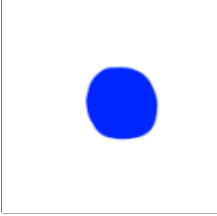
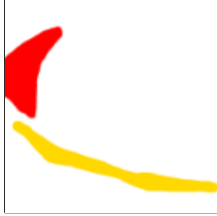
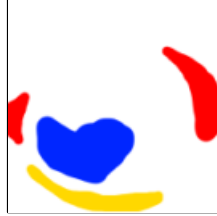
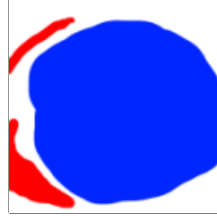
Input image					
Input mask					
Disease	Cherry Eye	Glaucoma	Uveitis	Corneal ulceration	Bacterial Keratitis
Symptom(s)	Symptom 1	Symptom 2	Symptom 3 Symptom 4	Symptom 2 Symptom 3 Symptom 4	Symptom 2 Symptom 3

Fig. 5. A test dataset sample consisting of 25 cases describing 5 disease in equal appearances. Symptom 1 stands for a colored mass protrusion in the corner of the eye, Symptom 2 for cloudiness or haziness of the cornea, Symptom 3 for sclera redness and Symptom 4 for excessive tearing.

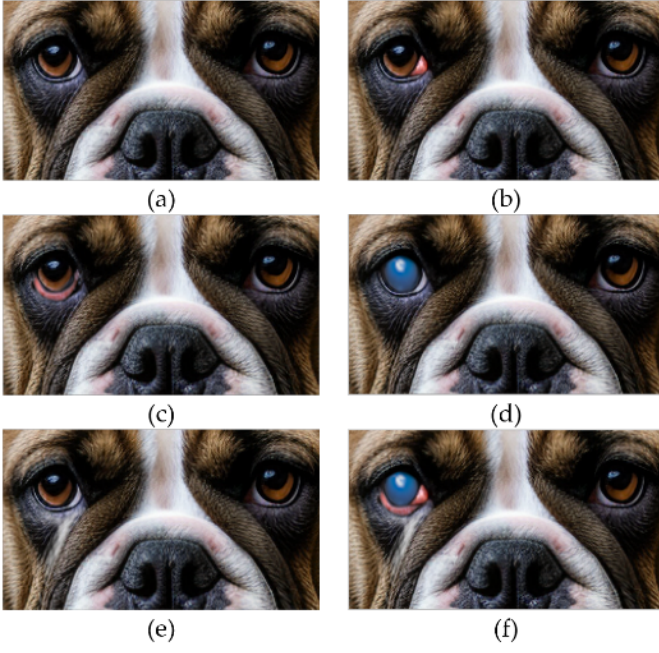


Fig. 6. Example of images containing symptoms generated with Stable Diffusion. From top left to bottom right: (a) healthy eyes, (b) left eye with prolapsed eyelid gland, (c) left eye with red sclera, (d) left eye with cloudy cornea, (e) left eye with epiphora and (f) eye with all symptoms.

The final dataset consists of diverse images representing different breeds and varying severities of each condition as seen in Fig. 7. This diversity ensures robustness in training machine learning models for ophthalmic image analysis.



Fig. 7. Close-up images describing various diseases generated using Stable Diffusion and custom LoRA based on a subset of real images.

C. U-Net Convolutional Neural Network Implementation

The enhanced U-Net architecture, depicted in Fig. 9, incorporates ResNet as the backbone for its encoder and decoder blocks. This architecture is designed to improve feature extraction and reconstruction capabilities.

The encoder section comprises several encoder blocks (EB), which progressively downscale the input image while capturing essential features. Each EB in the encoder utilizes ResNet layers to leverage residual learning, enhancing the feature extraction process. The extracted features are then passed to the decoder section, which consists of decoder blocks (DB). The decoder blocks use upscaling mechanisms to reconstruct the output image from the extracted features. This combination of ResNet-enhanced encoder and decoder blocks aims to improve the accuracy and efficiency of the

image segmentation tasks.

A pre-trained U-Net network introduced in the [48] and specifically developed for ophthalmological canine image symptoms segmentation using ResNet34 as backbone trained on the real dataset of 200 images is taken as the base for evaluation and named according – U-Net(R).

To improve the U-Net(R) model's performance, additional training was conducted using a synthesized dataset generated by Stable Diffusion. The original dataset was expanded with an additional 400 images, 100 for each class, forming the combined dataset used for training the enhanced U-Net model, referred to as U-Net(RSD) as seen in Fig. 8.

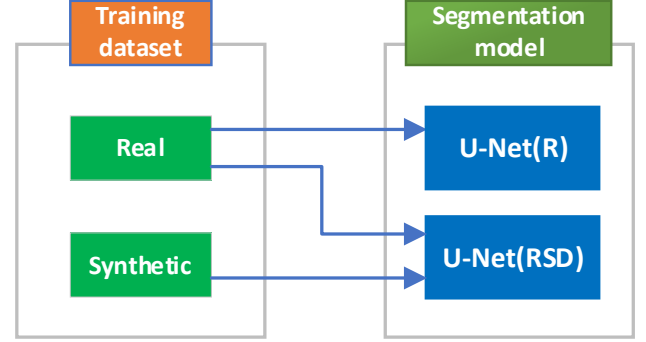


Fig. 8. Training datasets used for building U-Net(R) model contains only real images where U-Net(RSD) contain combination of Real and Synthetic images with corresponding masks.

D. GPT-4 and GPT-4o Analysis Approach

The GPT-4 model was utilized distinctly due to its nature as a language model rather than an image-processing tool. In this experiment, images are used as input to GPT-4 instead of a text sequence, so that a generated output corresponds to the visual content provided.

Current GPT-4 capabilities also offer image classification but lack the capability of object detection or segmentation. The next iteration, GPT-4o, handles object detection and segmentation and was used to request segmentation masks constructed in the same manner as the ground truth of test images.

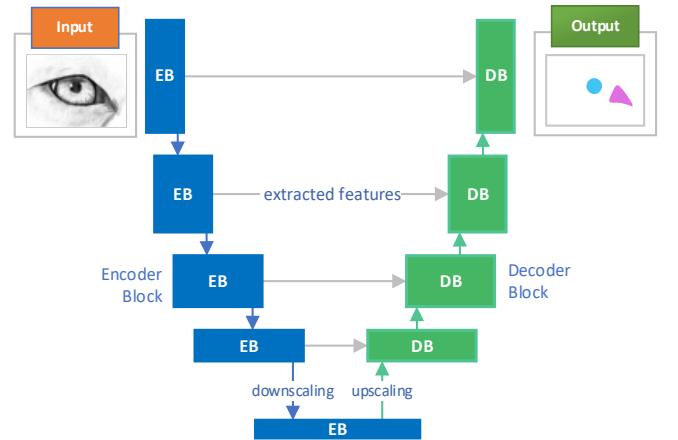


Fig. 9. Schematic of the enhanced U-Net architecture with ResNet backbone in both encoder and decoder blocks. The encoder blocks (EB) downscale the input while extracting features, and the decoder blocks (DB) upscale these features to reconstruct the output image.

GPT models, such as GPT-4, utilize a decoder structure based on the Transformer architecture designed for language generation tasks as shown in Fig. 10. The architecture comprises several key components:

1. **Embedding Layer:** The input tokens are first converted into embeddings, and positional encodings are added to these embeddings to retain information about the order of tokens.
2. **The Multi-Head Attention mechanism** allows the model to focus on different parts of the input sequence simultaneously, enhancing its ability to understand contextual relationships.
3. **Feed-Forward Network** that is position-wise and processes the output from the attention mechanism.
4. **Residual Connections and Layer Normalization** to facilitate better gradient flow and model convergence.
5. **Linear layer followed by a softmax layer** to generate, based on processed embeddings, probabilities for the next token in the sequence.

Based on descriptive image interpretation, both GPT-4 and

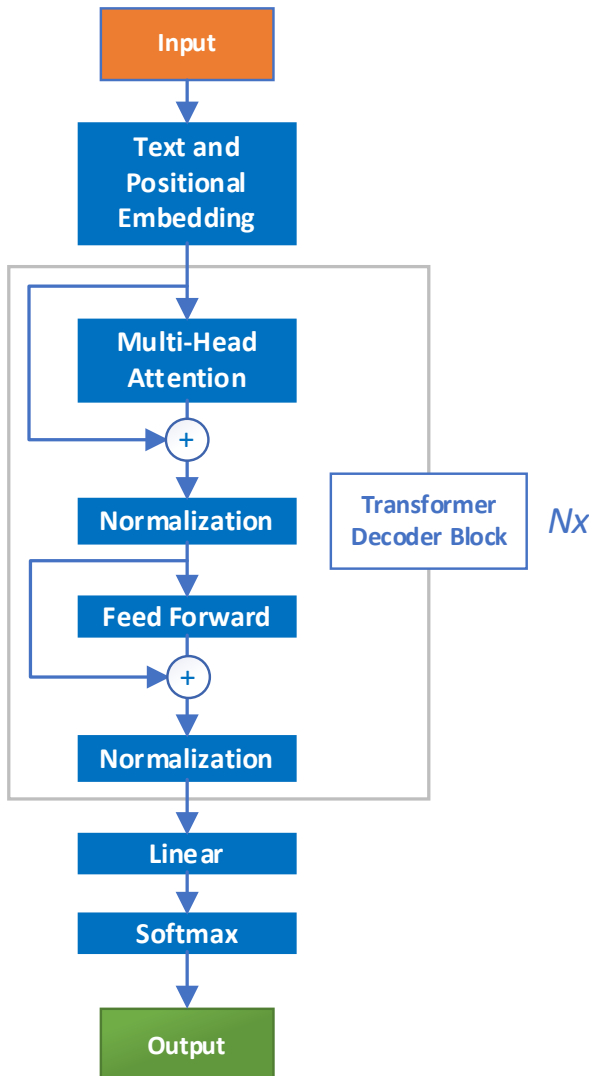


Fig. 10. The architecture of a Transformer Decoder block used in the GPT model. The model begins with text and positional embeddings of the input tokens. It then processes the embeddings through multiple layers of multi-head attention and feed-forward networks, each followed by normalization. The final output is generated after passing through a linear layer and a softmax function. The architecture allows the model to focus on different parts of the input sequence simultaneously, enhancing its contextual understanding and prediction capabilities.

GPT-4o can provide coherent responses that incorporate in-depth knowledge of veterinary ophthalmology. To return desirable results, these language models require context, which was engineered before requesting image analysis. The context prompt positions the model as a theoretical veterinary consultant tasked with observing an image and providing an educated guess on the most probable disease. The models are instructed to provide brief and relevant responses without exaggeration. Additionally, it is indicated that the segmentation must be made from the provided image without any background information. An example of the prompt is shown in Fig. 11:

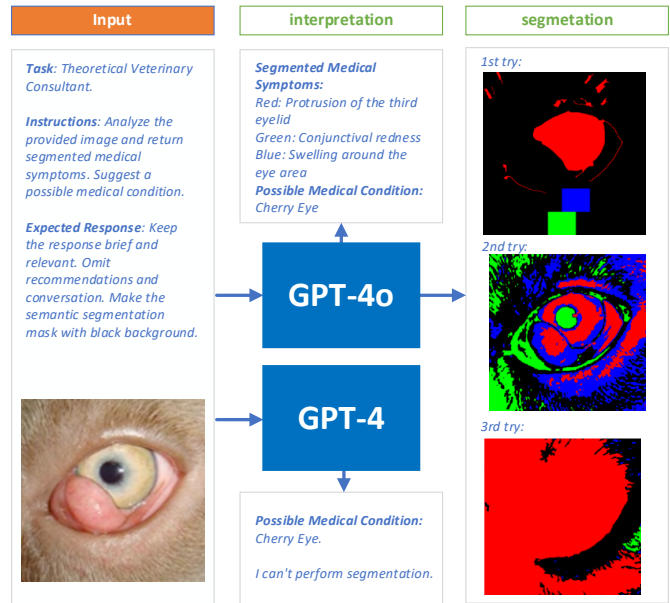


Fig. 11. An example of the prompt towards GPT-4 and GPT-4o with a request to segment medical symptoms and provide most probable diagnosis.

Results consisting of segmentation masks and diagnoses are collected and used in comparison with other methods.

E. Grounding SAM

The integration of Grounding Dino with SAM, so called Grounding SAM involves two key processes: object detection and image segmentation. Grounding DINO and Grounding SAM operate by integrating a transformer-based model with image recognition capabilities. The transformer model utilizes self-attention mechanisms to process and understand the relationships between different elements in an image. This mechanism allows the model to focus on relevant parts of the image when generating segmentation masks. By combining

this with zero-shot learning, Grounding DINO can detect

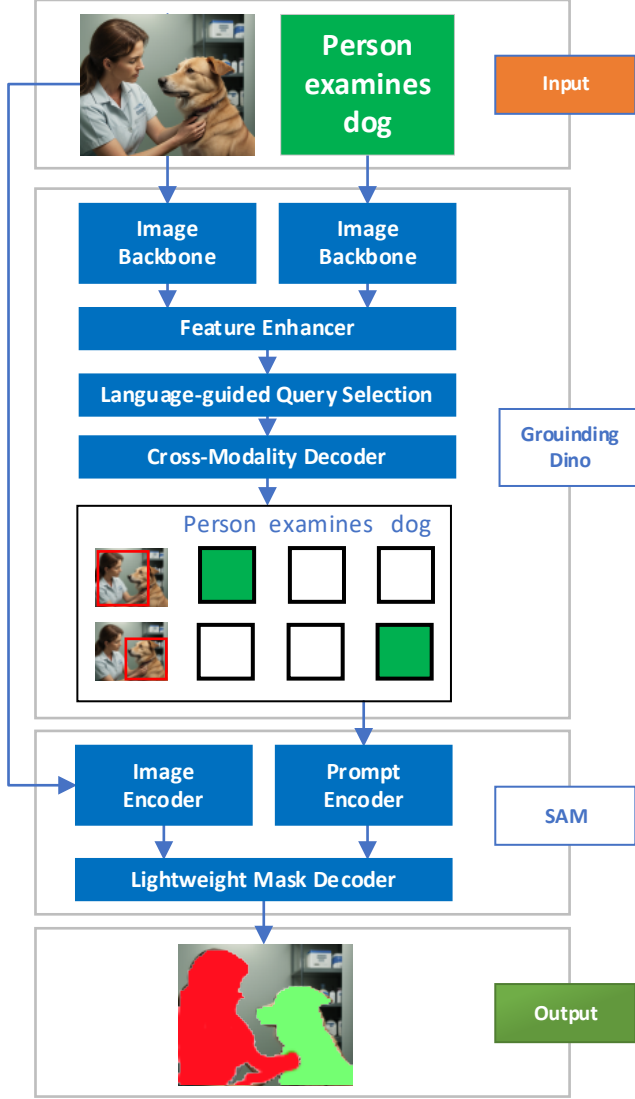


Fig. 12. Diagram illustrating the integration of Grounding Dino and SAM. The process begins with an input image, followed by feature extraction and object detection in Grounding Dino, and concludes with instance segmentation in SAM.

objects without prior training on specific datasets. In Grounding SAM detected objects' bounding boxes are then passed to SAM for instance segmentation, accurately outlining each identified object. This is achieved by using a large pre-trained model that can generalize across various tasks, converting bounding boxes into precise segmentation masks.

Inputs to Grounding Dino include images from the test dataset accompanied by text prompts describing symptoms in non-medical terms, such as "extensive tearing" instead of "epiphora", due to Grounding Dino's limited understanding of medical terminology. The output of Grounding Dino depends on a prompt and returns a bounding box around an object. The prompt is adjusted accordingly, and the presence is determined from the image observation.

The integration process is depicted in the diagram in Fig. 12., showing how the process begins with an input image depicting a person examining a dog. Grounding Dino first processes the input image through an image backbone to extract features, which are then further refined by a feature

enhancer. Using text prompts, the language-guided query selection guides the object detection process. The cross-modality decoder combines image and text features to detect objects, producing bounding boxes around the detected objects, such as the person and the dog. Next, these detected objects are processed by SAM, starting with the image encoder that encodes the input image for segmentation. The prompt encoder then encodes the text prompt to assist in segmentation. Finally, the lightweight mask decoder uses the encoded image and prompt to generate segmentation masks for the detected objects.

The ongoing experiment shows high sensitivity to a prompt structure and difficulties when medical terms are used therefore prompt engineering requires a higher level of attention. See Fig. 13 for an example.

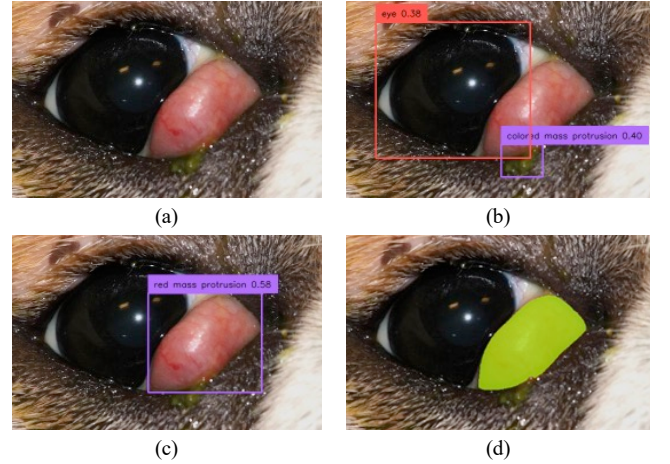


Fig. 13. Example of Grounding Dino prompting diversity with resulting detection and its prediction score. Prompts used for generating detection from top to bottom are: (a) "colored mass protrusion in the corner of the eye", (b) "colored mass protrusion in the eye" and (c) "red mass protrusion". Based on the third prompt segmentation using SAM is performed where bounding box is taken as a reference (d).

F. The proposed pipeline for generating canine ophthalmic diagnoses based on images of eye medical symptoms

The proposed pipeline is presented in the diagram in Fig. 14 where the additional information of particular medical symptoms of dog eye in terms of position and type are provided by the semantic segmentation model and added to an input image of dog eye as a prompt to multimodal LLM such as GPT-4 or GPT-4o to get as accurate diagnostic interpretation of symptoms depicted in image as possible. The most favorable procedure that requires the least involvement from the user and gives the best results in terms of diagnosis accuracy is highlighted with a red arrow.

In the pipeline is also presented a possible simplified scenario of using LLM for diagnosis based on the input image of the diseased eye which does not include image enhancement with a segmented mask. There is also a variant that does not include the use of a customized model for the segmentation of eye diseases such as U-Net, but transformer models that are capable of zero-shot learning like Grounding SAM, but need appropriate explanation with the image that describes what should be segmented in the image.

Each of the elements of the pipeline system is described in detail in the experimental part and evaluated qualitatively and quantitatively with appropriate metrics.

IV. EXPERIMENT

The experiment is divided into the segmentation of symptoms and the interpretation of input images with and without segmentation masks from the most successful method. The results of both parts of the experiment will be evaluated using general metrics for the segmentation and text similarity compared to ground truth. The metrics and procedures of the experiment will be thoroughly explained in the subsections.

A. Segmentation evaluation metrics

The performance of segmentation models was evaluated using the next metrics:

- **Jaccard Index**, also known as the Intersection-over-union (IoU) metric measures the overlap between the predicted segmentation and the ground truth segmentation, calculated as follows:

$$\text{IoU} = \frac{TP}{TP + FP + FN} \quad (1)$$

where TP represents true positives (correctly identified positive cases), FP false positives (incorrectly identified positive cases), and FN false negatives (missed positive cases).

- **Dice Index**, Similar to the Jaccard Index, this metric gives more weight to the overlap between the predicted and ground truth segmentations, calculated as follows:

$$\text{Dice Index} = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (2)$$

- **Pixel Accuracy** metric calculates the proportion of correctly classified pixels in the entire image, calculated as follows:

$$\text{Pixel Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

where TN represents true negatives (correctly identified negative cases). The Pixel Accuracy is often used alongside other metrics like Sensitivity and Specificity to provide a comprehensive evaluation.

- **Sensitivity**, also known as the true positive rate or recall, measures the proportion of actual positives that are correctly identified as such. It is calculated as follows:

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (4)$$

- **Specificity**, also known as the true negative rate, measures the proportion of actual negatives that are correctly identified. It is calculated as follows:

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (5)$$

It is important to distinguish Sensitivity and Specificity from Precision and Recall used in object detection. Precision is the proportion of positive identifications that are actually correct, calculated as the number of true positives divided by the sum of true positives and false positives. Recall, in the context of object detection, is synonymous with Sensitivity. For all mentioned metrics a higher result indicates better performance of the segmentation model, with a result of 1 representing perfect match of ground truth and predicted mask.

B. Text similarity evaluation metrics

The performance of GPT-4 and GPT-4o in generating textual diagnoses was evaluated using the following metrics: BERTScore, CLIPScore, BLEU, METEOR, ROUGE, and SPICE. These metrics were chosen to effectively measure the nuances in language comprehension necessary for precise medical diagnoses.

- **BERTScore** employs BERT, a pre-trained language model, to create contextual embeddings for each token in a sentence. The cosine similarity between the candidate and reference tokens is calculated as follows:

$$\text{BERTScore}(P_B, R_B, F_B) = P, R, F1 \quad (6)$$

The formulas for Precision (P_B) and Recall (R_B) are:

$$P_B = \frac{1}{|C|} \sum_{t \in C} \max_{t' \in R} \cos(\text{emb}(t), \text{emb}(t')) \quad (7)$$

$$R_B = \frac{1}{|R|} \sum_{t' \in R} \max_{t \in C} \cos(\text{emb}(t'), \text{emb}(t)) \quad (8)$$

where C is the set of tokens in the candidate sentence, R is the set of tokens in the reference sentence, and $\cos(\text{emb}(t), \text{emb}(t'))$ represents the cosine similarity between the embeddings of tokens t and t'.

The F1 Score (F_B) is calculated as:

$$F_B = 2 \frac{P_B R_B}{P_B + R_B} \quad (9)$$

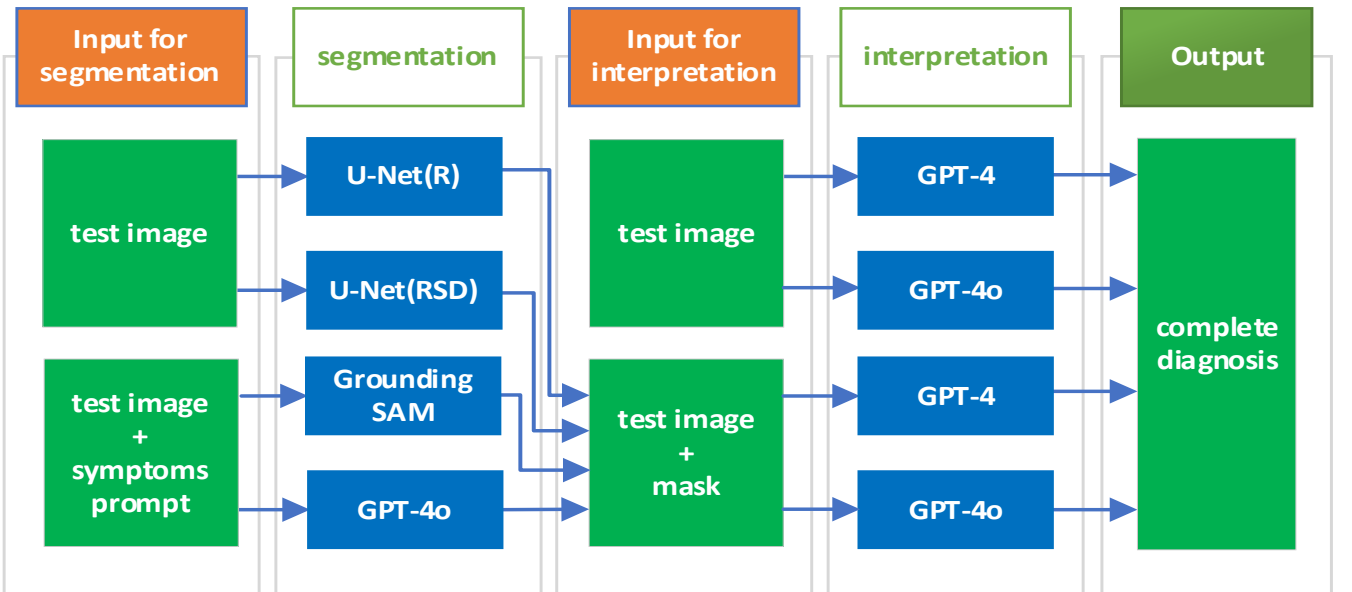


Fig. 14. The proposed pipeline for generating canine ophthalmic diagnoses based on image of eye medical symptoms.

- **CLIPScore** evaluates text by creating vector representations using CLIP, a multimodal vision and language model, and calculates the cosine similarity between these vectors and the reference text:

$$CLIPScore = \frac{V_{candidate} V_{reference}}{\|V_{candidate}\| \|V_{reference}\|} \quad (10)$$

where $V_{candidate}$ and $V_{reference}$ are the vector representations of the candidate and reference texts respectively.

- **BLEU** assesses the geometric mean of n-gram precision in matching candidate and reference descriptions, with a brevity penalty to prevent overly short sentences:

$$BLEU = BP \cdot \exp \left(\sum_{n=1}^N w_n \log p_n \right) \quad (11)$$

where BP is the brevity penalty, w_n are weights, and p_n is the precision of n-grams.

- **METEOR** computes the harmonic mean of unigram precision and recall of matches between candidates and references, leveraging stemmers, WordNet synonyms, and paraphrase tables:

$$METEOR = F_{METEOR} = 10 \frac{PR}{9P + R} \quad (12)$$

where P is precision and R is recall.

- **ROUGE** identifies the longest common subsequence of tokens between a candidate and a reference without requiring consecutive matches. One of its variants, ROUGE_L, is calculated as:

$$ROUGE_L = \frac{(1 + \beta^2)RP}{R + \beta^2 P} \quad (13)$$

where β is a weighting factor, R is recall, and P is precision.

- **SPICE** evaluates image captions by comparing scene graphs of candidate and reference sentences based on their semantic propositional content:

$$SPICE = F_{SPICE} = 2 \frac{PR}{P + R} \quad (14)$$

where P and R are precision and recall calculated from the scene graph tuples.

These metrics provide a thorough evaluation of the LLMs' ability to interpret medical symptoms, ensuring the robustness and reliability of the preliminary diagnosis support these models offer.

C. Segmentation

Segmentation is the default output of U-Net, a convolutional neural network architecture commonly used for biomedical image segmentation. In this study, two U-Net models, U-Net(R) and U-Net(RSD), will be evaluated using a test dataset containing 25 images divided into 5 disease groups. Each group represents a different type of pathology, providing a comprehensive assessment of the models' performance across varied conditions. Grounding SAM provides an initial segmentation based on the provided bounding boxes, enabling a comparative analysis against the U-Net models. The recently introduced features of GPT-4o include advanced capabilities in object detection and semantic segmentation, so GPT-4o can also be used as a candidate for segmentation evaluation.

The results of the evaluation are presented in the Table I where it can be observed that the most successful method

using Pixel Accuracy, IoU and Dice Index metric is the U-Net(RSD).

Further assessments of U-Net(RSD) model related to particular class of diagnosis and additional metrics are

TABLE I
EVALUATION OF U-NET, U-NET(RSD), GPT-4o AND GROUNDING SAM SEGMENTATION USING JACCARD (IoU), DICE INDEX AND PIXEL ACCURACY METRICS.

Method	IoU Index	Dice Index	Pixel Accuracy
U-Net(R)	0.74	0.83	0.94
U-Net(RSD)	0.81	0.88	0.97
GPT-4o	0.16	0.20	0.72
Grounding SAM	0.14	0.16	0.67

presented in Table II.

It can be observed that S4 - "colored mass protrusion in the

TABLE II
EVALUATION OF U-NET(RSD) USING IoU AND DICE INDEX WITH STANDARD DEVIATION AND PIXEL ACCURACY, SENSITIVITY AND SPECIFICITY ON INDIVIDUAL SYMPTOM: S1 - CLOUDINESS OR HAZINESS OF THE CORNEA; S2 - SCLERA REDNESS; S3 - EXCESSIVE TEARS; S4 - COLORED MASS PROTRUSION IN THE CORNER OF THE EYE.

	IoU Index (mean $\pm$ std)	Dice Index (mean $\pm$ std)	Pixel Accuracy	Sensitivity	Specificity
S1	0.84 $\pm$ 0.08	0.91 $\pm$ 0.04	0.991	0.944	0.972
S2	0.74 $\pm$ 0.20	0.83 $\pm$ 0.15	0.976	0.928	0.973
S3	0.74 $\pm$ 0.26	0.82 $\pm$ 0.20	0.972	0.851	0.983
S4	0.90 $\pm$ 0.22	0.92 $\pm$ 0.20	0.984	0.932	0.991

corner of the eye" symptom has highest scores by Specificity, IoU and Dice Index. The most difficult symptom to be segmented is S3 - "excessive tearing".

In the Fig. 15 Dice Index is visualized for each symptom using different method. U-Net(RSD) has the highest score in three out of four symptoms. The second best and the one with the highest Dice Index score for "colored mass protrusion in the corner of the eye" is U-NET(R). The GPT-4o showed higher score with symptoms which are usually more scattered and irregularly shaped than those which are clustered.

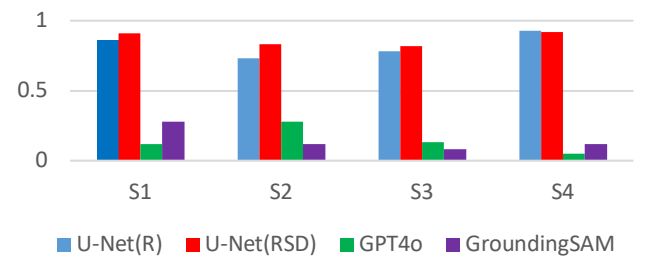


Fig. 15. Visual presentation of Dice Index for a different segmentation methods considering four different classes.

D. Interpretation using LLM

Since U-Net and Grounding SAM methods are designed for semantic image segmentation and not for textual interpretation, this section evaluates GPT-4 and GPT-4o models that can generate textual diagnoses based on input images. In this experiment, GPT-4 and GPT-4o will be prompted to provide the two most probable medical diagnoses, both with and without the mask generated from U-Net(RSD) segmentation. The robustness and reliability of making preliminary medical diagnoses generated by these approaches are evaluated using selected text similarity metrics. In Table III evaluation results in case of single and dual medical diagnosis of GPT-4 and GPT-4o with and without masks generated using U-Net(RSD) segmentation are shown. In both dual and single diagnosis cases the results of all metrics are in favor of GPT-4o when segmentation masks are provided. The second-best results are when GPT-4o is used with the input image alone. In case when GPT-4 is used in a single diagnosis results support the usage of mask input opposite to dual diagnosis where better results are if

TABLE III
EVALUATION OF GPT-4 AND GPT-4o WITH AND WITHOUT MASKS
GENERATED USING U-Net(RSD) SEGMENTATION IN CASE OF SINGLE AND
DUAL MEDICAL DIAGNOSIS

	BERT Score	CLIP Score	BLEU	MET.	ROUGE	SPICE
single diagnosis						
GPT-4o image	0.83	0.90	0.48	0.31	0.48	0.48
with mask	0.84	0.91	0.49	0.32	0.51	0.51
GPT-4 image	0.63	0.84	0.08	0.06	0.08	0.08
with mask	0.65	0.86	0.15	0.08	0.16	0.15
dual diagnosis						
GPT-4o image	0.62	0.90	0.14	0.13	0.17	0.45
with mask	0.64	0.91	0.16	0.14	0.19	0.49
GPT-4 image	0.59	0.85	0.07	0.05	0.08	0.11
with mask	0.59	0.85	0.03	0.03	0.04	0.06

masks are not provided.

Performance of GPT-4 vs. GPT-4o in determining disease type is demonstrated in Fig. 16. In four out of five diseases GPT-4o shows superiority. It is especially successful in resolving every case of Uveitis and 80% of Cherry Eye. The only occurrence when GPT-4 outperforms GPT-4o is in

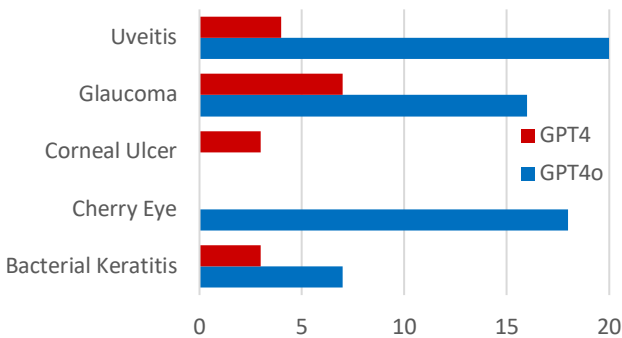


Fig. 16. The difference between GPT-4 and GPT-4o in diagnosis of four canine eye diseases.

diagnosing Corneal Ulcer, which GPT-4o was unable to predict at all.

The diagram in Fig. 17. shows the overall results of individual medical cases in a logarithmic scale for GPT-4o in single and dual diagnosis. The values further away from the center correspond to higher similarity, described by BERTScore, in diagnosis. Results of the single diagnosis generally have higher scores, especially in four out of five Cherry Eye cases. Results are slightly in favor of dual diagnosis in cases of Bacterial Keratitis and Corneal Ulcer.

V. DISCUSSION

The experimental results demonstrate that segmentation performance using U-Net improves with the addition of a custom training dataset. Real canine ophthalmological cases are challenging to obtain and require substantial resources. Until recently, generating synthetic datasets was time-consuming, resource-intensive, and required extensive expertise to produce images of acceptable quality. The introduction of Stable Diffusion has significantly eased these challenges, enabling the generation of high-quality synthetic images in a shorter time and at a lower cost. Although these computer-generated images still need expert validation to confirm symptoms, the process can be automated. This methodology can be applied in other medical and non-medical fields, with the assumption that it will improve training outcomes using U-Net and potentially other CNN-based methods.

The segmentation results using GPT-4o are not as favorable as those achieved with U-Net(R) or U-Net(RSD). However, given that this technology is continually improving with each release, it is reasonable to expect that GPT-4o will narrow this performance gap in the future. Grounding SAM, the most recent approach shows generally good results but lacks medical knowledge, heavily relying on prompts, leading to significant variation in outcomes. With further research, it is reasonable to expect substantial improvements

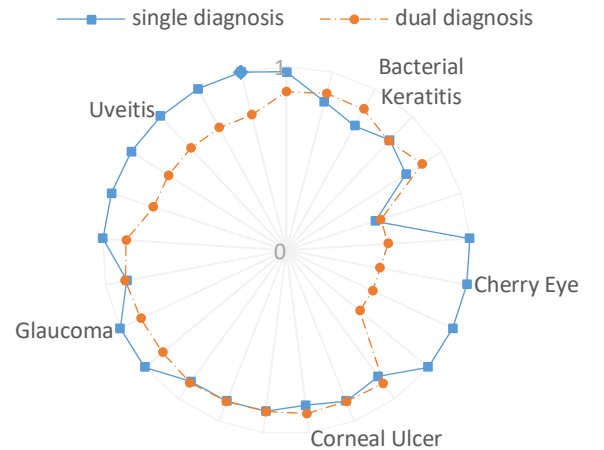


Fig. 17. BERTScore results for each case divided into four disease groups when single and dual diagnoses approach is applied.

in this method as well.

A notable disadvantage of U-Net segmentation compared to GPT-4o, and Grounding SAM is the necessity of a new

training dataset whenever a new symptom is introduced. In contrast, the generalization capabilities of Grounding SAM and especially GPT-4o provide flexibility and can be quickly adapted to new challenges. The use of GPT-4 and GPT-4o has proven to be a good choice for canine ophthalmological diagnosis, with cases being correctly predicted within a single diagnosis attempt. Dual diagnosis shows even better accuracy in predicting diseases, although these results are harder to quantify using the metrics described in this paper. While additional diagnoses lower certainty, as shown in Table III, they provide valuable information to veterinary specialists and concerned pet owners.

The described automatic diagnosis procedure, which consists of segmented images and disease prediction, is promising, but still has room for improvement, which will be the focus of further research. A significant limitation of the current method is its strong dependence on the intellectual property of the LLMs used. To make this approach independent, it is suggested to use models that can be run locally with full control. However, this presents challenges due to the size of the model and the resources required to process requests.

VI. CONCLUSION

This study proposes a pipeline for the automatic generation of canine ophthalmology diagnoses based on the image of a dog eye as shown in Fig. 18. The pipeline also assumes the additional use of a model for segmentation of the input image to highlight key information in the image related to the location and type of medical symptoms to provide the GPT model for the interpretation of diagnoses with additional information that can favorably influence the increase of its accuracy.

For every part of the pipeline, a comprehensive evaluation is performed. For the segmentation part U-Net models, one fine-tuned on real medical images and the other on a mixture of real and synthetic images are compared with Grounding

SAM and GPT-4o model that are zero-shot segmentation models guided by textual description. To generate diagnoses of the diseased eye, the results of GPT-4 and GPT-4o were analyzed for the case when they were given an image with or without an additional segmentation mask.

We started with the assumption that integrating these advanced computer vision techniques in one pipeline would improve diagnostic accuracy and efficiency in veterinary ophthalmology compared to using these methodologies independently. The findings confirm this, demonstrating that each methodology offers unique strengths and suffer from weaknesses, but certain combinations can enhance diagnostic workflows.

The enhanced U-Net(RSD) model fine-tuned on mixture of real and simulated image symptoms excelled in image segmentation tasks, precisely delineating ocular structures and proving robust for segmentation purposes. However, its limitation lies in its inability to provide interpretative analysis, which restricts its standalone application in comprehensive diagnostic workflows. GPT-4o, on the other hand, generated detailed interpretative textual descriptions akin to a specialist's assessment. Despite its interpretative strengths, GPT-4o's segmentation capabilities were less refined and inconsistent than those of the U-Net model. Grounding SAM shows very good semantic segmentation potential, but its development stage is less mature than U-Net, resulting in variability in performance due to reliance on prompt engineering and limited medical terminology understanding.

Future research should focus on using the proposed pipeline that combines the high segmentation accuracy of the U-Net model with the interpretive capabilities of the GPT-4o method and exploits their complementary strengths in the best possible way to create a comprehensive diagnostic tool.

Further advancements in Grounding SAM, particularly in its understanding of medical terminology, could enhance its overall performance and reliability. Additionally, exploring

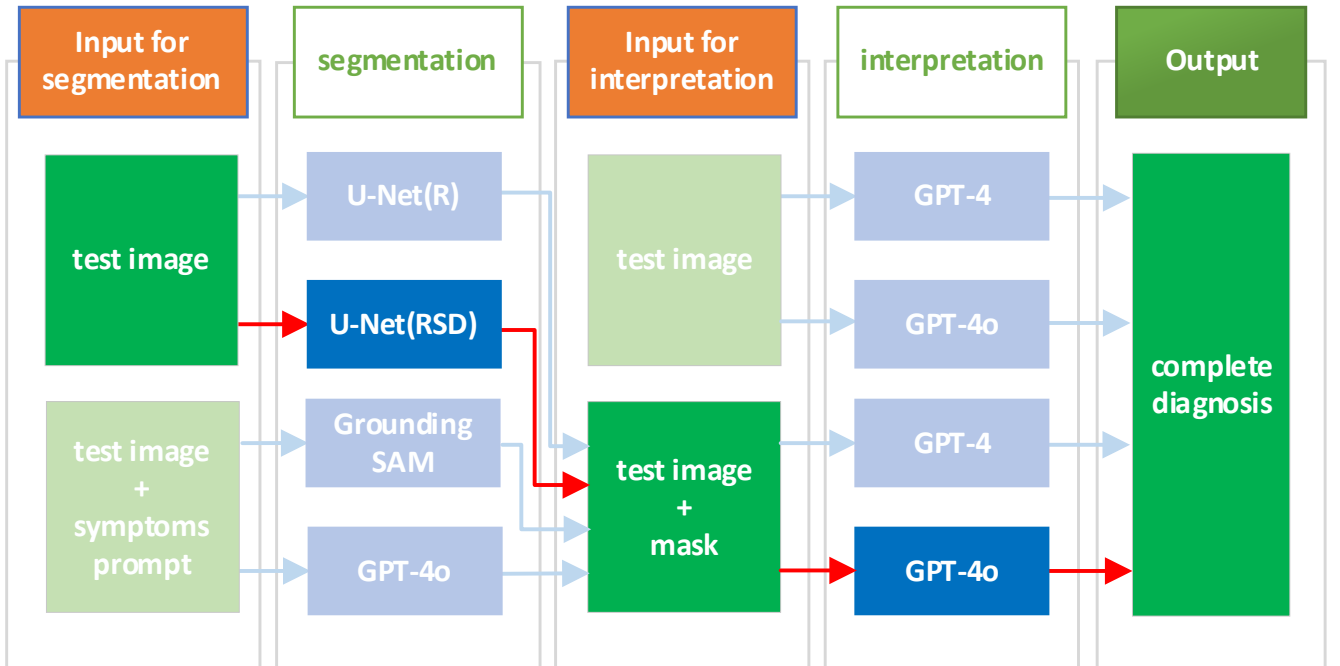


Fig. 18. The evaluated pipeline for generating canine ophthalmic diagnoses based on image of eye medical symptoms showing best results when segmented mask images using U-Net(RSD) are used as input along with input image into GPT-4o.

the potential of incorporating high-quality synthetic datasets generated by models like Stable Diffusion could improve the robustness and generalizability of segmentation models.

Efforts should also address the current limitations of each methodology, including improving the interpretative capabilities of segmentation models and enhancing the segmentation accuracy of language models. By focusing on these areas, the integration of advanced computer vision techniques can significantly enhance diagnostic accuracy and efficiency in veterinary ophthalmology, ultimately improving clinical outcomes and patient care.

In conclusion, this research highlights the potential benefits and limitations of advanced computer vision methodologies in veterinary ophthalmology. The findings suggest that a synergistic approach, leveraging the strengths of each methodology, could pave the way for significant innovations in the field. Future research should aim to refine these techniques and explore their broader applicability in both veterinary and medical imaging. The promising results of this study underscore the transformative potential of integrating multiple advanced computer vision technologies to advance veterinary diagnostics.

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